

Problem Set for 13 March 2025

Exercise 1 Let $f : U \rightarrow V$ be linear map and $\dim U = n, \dim V = m$.

1. Let $E \subset V$ be an k -dimensional subspace, prove that

$$\dim f^{-1}(E) \geq n - m + k.$$

答: 考虑映射

$$f_E : f^{-1}(E) \rightarrow V, \quad u \mapsto f(u)$$

此时

$$\begin{aligned} \dim f^{-1}(E) & \stackrel{\text{rank-null}}{=} \dim \ker f_E + \dim \operatorname{im} f_E \\ & \stackrel{\ker f = \ker f_E}{=} \dim \ker f + \dim \operatorname{im} f_E \\ & \stackrel{\text{rank-null}}{=} (\dim U - \dim \operatorname{im} f) + \dim \operatorname{im} f_E \\ & \stackrel{*}{\geq} (\dim U - \dim V) + \dim E. \end{aligned}$$

特别提示 $\operatorname{im}(f_E)$ 是 E 的子空间 (不必取等), 因为 f 未必是满射.

最后* 解释如下: 将 $\operatorname{im}(f_E)$ 扩充成 E 的子空间, 需要额外添加线性无关组 $\mathcal{B} \subset V \setminus \operatorname{im}(f)$. 使用该 $\mathcal{B}$ 扩充 $\operatorname{im}(f)$, 得 V 的子空间. 从而

$$\dim V - \dim \operatorname{im}(f) \geq \dim E - \dim \operatorname{im}(f_E).$$

2. When f is surjective, the equality holds.

答: f 满时,

$$\dim V - \dim \operatorname{im}(f) = 0 = \dim E - \dim \operatorname{im}(f_E).$$